The documentation and process conversion measures necessary to comply with this revision shall be completed by .15 August 1999

INCH-POUND

MIL-PRF-19500/511C
15 May 1999
SUPERSEDING
MIL-S-19500/511B
24 June 1994

PERFORMANCE SPECIFICATION SHEET
SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, SWITCHING
TYPE 2N4261 JAN, JANTX, JANTXV AND JANS

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for PNP silicon, switching transistors. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (similar toTO-72).

1.3 Maximum ratings.

P_T T_A = +25°C 1/	V_{CBO}	V_{CEO}	V_{EBO}	I_C	T_J	T_J and T_{STG}	$R_{\theta JA}$
MW	V dc	V dc	V dc	mA dc	°C	°C	°C/mW
200	15	15	4.5	30	200	-65 to +200	0.860

1/ Derate linearly 2.3 mW/°C Between T_A = +25°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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1.4 Primary electrical characteristics at T_A = +25°C, unless otherwise specified.

	h_{FE1} 1/	h_{FE2} 1/	h_{FE3} 1/	$\|h_{fe}\|$	r' b' C_{cl}	Switching	
	V_{CE} = 1.0 V dc I_C = 1 mA dc	V_{CE} = 1.0 V dc I_C = 10 mA dc	V_{CE} = 1.0 V dc I_C = 30 mA dc	V_{CE} = 10 V dc I_C = 10 mA dc f = 100 MHz	V_{CE} = 4.0 V dc I_C = 5 mA dc f = 31.8 MHz	t_{on}	t_{off}
					ps	Ns	ns
Min	25	30	20	20			
Max		150			60	2.5	3.5

	$V_{CE(sat)1}$ I_C = 1 mA dc I_B = 0.1 mA dc	$V_{CE(sat)2}$ I_C = 10 mA dc I_B = 1.0 mA dc	$V_{BE(sat)1}$ I_C = 1 mA dc V_{CE} = 1 V dc	$V_{BE(sat)2}$ I_C = 10 mA dc V_{CE} = 1 V dc	C_{obo} V_{CB} = 4 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz
	V dc	V dc	V dc	V dc	PF
Min					
Max	0.15	0.35	0.8	1.0	2.5

1/ Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

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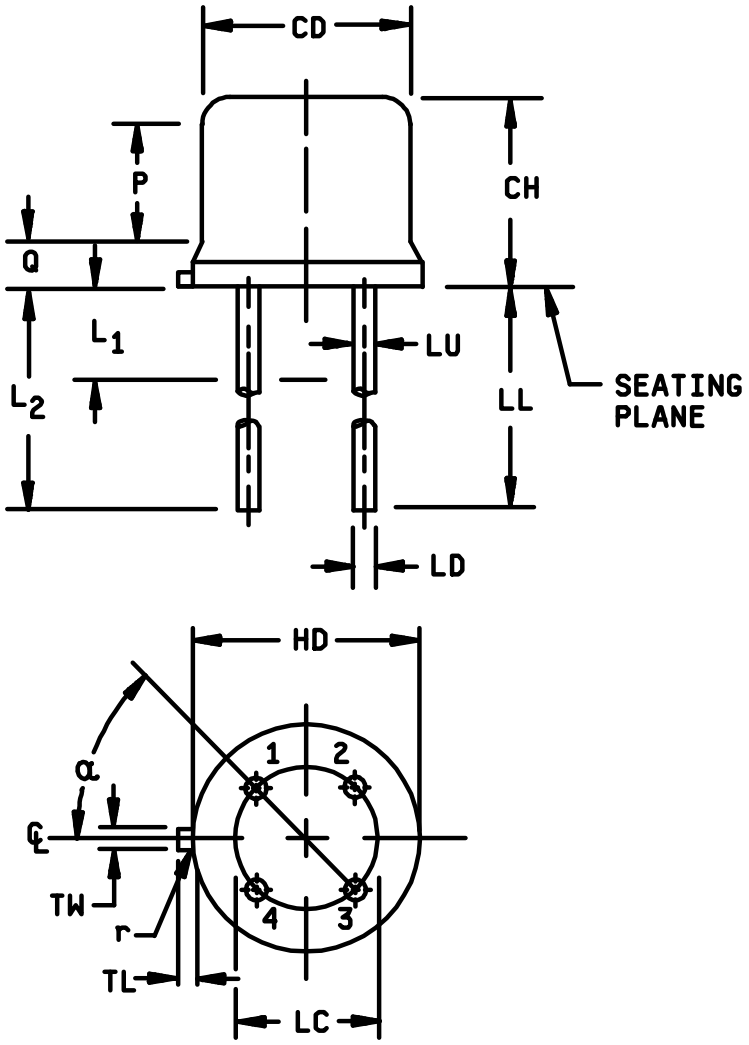


FIGURE 1. Physical dimensions (similar to TO-72).

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Ltr	Dimensions				Notes
	Min	Max	Min	Max	
CD	0.178	0.195	4.52	4.95	6
CH	0.170	0.210	4.32	5.33	
HD	0.209	0.230	5.31	5.84	
LC	0.100 TP		2.54 TP		7
LD	0.016	0.021	0.41	0.53	8, 9
LL	0.500	0.750	12.7	19.05	8, 9
LU	0.016	0.019	0.41	0.48	8, 9
L_1		0.050		1.27	8, 9
L_2	0.250		6.35		8, 9
TL	0.028	0.048	0.71	1.22	4
TW	0.036	0.046	0.91	1.17	3
P	0.100		2.54		6
Q		0.04		1.02	
r		0.007		0.18	11
α	45° TP				7

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of 0.11 inch (0.028 mm).
4. TL measured from maximum HD.
5. Outline in this zone is not controlled.
6. CD shall not vary more than 0.010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
7. Leads at gauge plane 0.054 + 0.001 – 0.000 inch (1.37 + 0.03 – 0.00 mm) below seating plane shall be within 0.007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods .
8. LU applies between L_1 and L_2. LD applies between L_2 and LL minimum.
9. All three leads.
10. This lead shall be electrically connected to the case.
11. r (radius) applies to both inside corners of tab.
12. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions (similar to TO-72) continued.

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3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figure 1 (similar to TO-72) herein.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500. Where a choice of lead finish is desired, it shall be specified in the contract or purchase order (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3.

3.7 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2).

4.VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.3 Screening. Screening shall be in accordance with MIL-PRF-19500 (table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurements	
	JANS	JANTX, JANTXV levels
9	I_{CEX2} and h_{FE2}	Not applicable
11	I_{CEX2} and h_{FE2}; ΔI_{CEX2} = 100 percent of initial value or 1 nA dc, whichever is greater. Δh_{FE2} = ± 15 percent of initial value.	I_{CEX2} and h_{FE2}
12	See 4.3.1.	See 4.3.1.
13	Subgroup 2 and 3 of table I herein; ΔI_{CEX2} = 100 percent of initial value or 1 nA dc, whichever is greater. Δh_{FE2} = ± 25 percent of initial value.	Subgroup 2 of table I herein; ΔI_{CEX2} = 100 percent of initial value or 2 nA dc, whichever is greater. Δh_{FE2} = ± 25 percent of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_A = Room ambient in accordance with the general requirements of MIL-STD-750 (see 4.5). V_{CB} = 10 V dc, P_T = 200 mW.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein. Electrical measurements (end-points) shall be in accordance with table II herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 10 V dc; P_T = 200 mW, T_A = Room ambient in accordance with the general requirements of MIL-STD-750 (see 4.5), t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced air cooling on devices shall be permitted.
B5	1027	T_A = 125°C ± 25°C; for 96 hours; P_T = 200 mW at 100°C or adjusted as required by the chosen T_A to give an average lot T_J = 275°C. Marking legibility requirements shall not apply; V_{CB} = 10 V dc.

4.4.2.2 Group B inspection, table VIb (JAN, JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1027	V_{CB} = 10 V dc; P_T = 200 mW, T_A = Room ambient in accordance with the general requirements of MIL-STD-750 (see 4.5),
B5		Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

Subgroup	Method	Condition
C2	2036	Test condition E.
C6	1026	V_{CB} = 10 V dc; P_T = 200 mW, T_A = Room ambient in accordance with the general requirements of MIL-STD-750 (see 4.5),

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Collector base time constant. This parameter may be determined by applying an RF signal voltage of 0.5 volt (rms) across the collector base terminals, and measuring the ac voltage drop (V_{eb}) with a high impedance RF voltmeter across the emitter base terminals. With f = 31.8 MHz used for the 0.5 volt signal, the following computation applies: r'b, Cc: (ps) = 10 x Veb (millivolts).

4.5.3 Base cutoff current. This parameter shall be measured using method 3041, bias condition A, of MIL-STD-750. The value of the base to emitter cutoff current shall be read on the current meter inserted in the base circuit.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical Examination	2071					
Subgroup 2						
Collector to emitter Breakdown voltage	3011	Bias condition D, I_C = 10 mA dc;	$V_{(BR)CEO}$	15		V dc
Collector to base Breakdown voltage	3001	Bias condition D, I_C = 10 µA dc	$V_{(BR)CBO}$	15		V dc
Emitter-base cutoff current	3061	Bias condition D, V_{EB} = 4.5 V dc	I_{EBO}		10	µA dc
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = 0.4 V dc V_{CE} = 10 V dc	I_{CEX1}		50	nA dc
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = 2.0 V dc V_{CE} = 10 V dc	I_{CEX2}		5.0	nA dc
Base cutoff current		V_{BE} = 2.0 V dc; V_{CE} = 10 V dc, see 4.5.3	I_{BEX}		5.0	nA dc
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 1.0 mA dc; pulsed (see 4.5.1)	h_{FE1}	25		
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	h_{FE2}	30	150	
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 30 mA dc; pulsed (see 4.5.1)	h_{FE3}	20		
Base to emitter voltage (non-saturated)	3066	Test condition B; V_{CE} = 1.0 V dc; I_C = 1.0 mA dc;	V_{BE1}		0.8	V dc
Base to emitter voltage (non-saturated)	3066	Test condition B; V_{CE} = 1.0 V dc; I_C = 10 mA dc;	V_{BE2}		1.0	V dc
Collector to emitter saturated voltage	3071	I_C = 1.0 mA dc; I_B = 0.1 mA dc	$V_{CE(sat)1}$		0.15	V dc
Collector to emitter saturated voltage	3071	I_C = 10 mA dc; I_B = 1.0 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		0.35	V dc

See footnote at end of table.

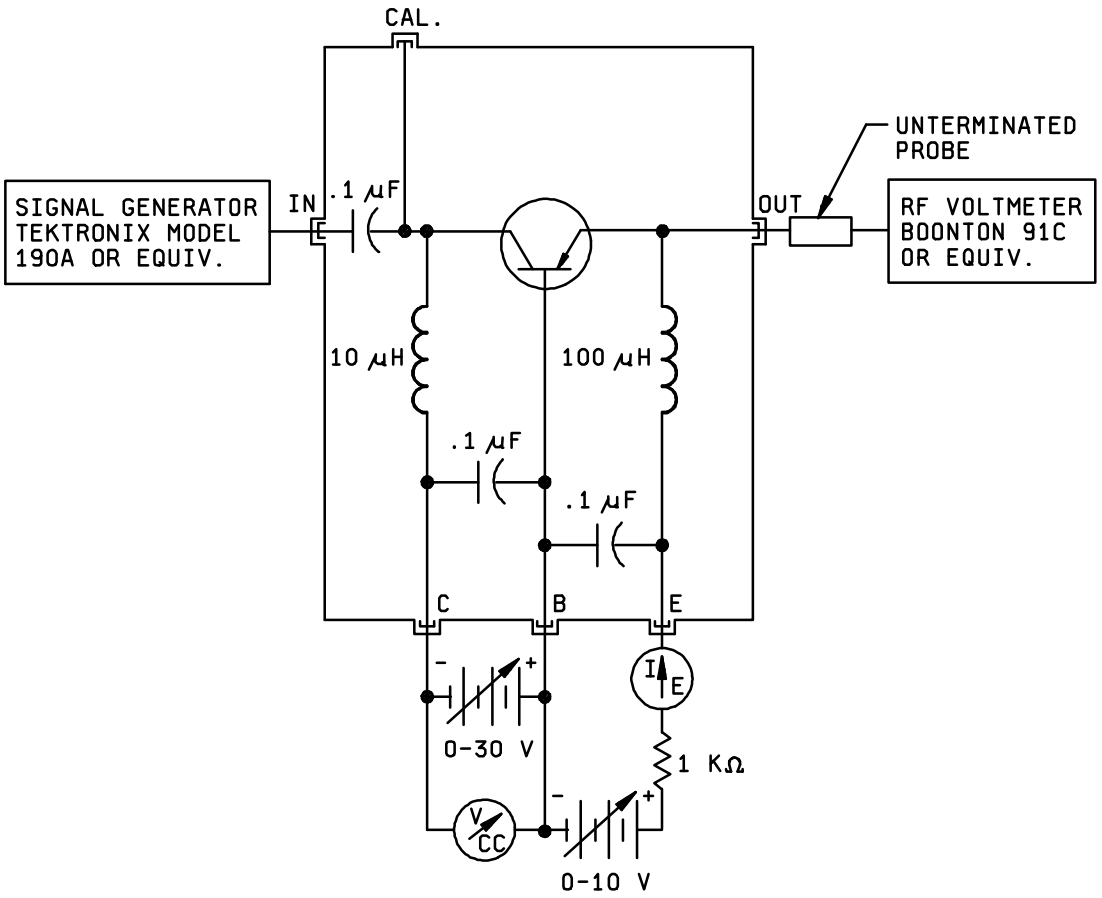
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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High-temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = 2.0 V dc V_{CE} = 10 V dc	I_{CEX3}		5.0	µA dc
Low-temperature operation :		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc I_C = 10 mA dc; pulsed (see 4.5.1)	h_{FE4}	15		
Subgroup 4						
Magnitude of common emitter small-signal short-circuit forward- current transfer ratio	3306	V_{CE} = 4.0 V dc; I_C = 5.0 mA dc; f = 100 MHz	$\|h_{fe1}\|$	15		
Magnitude of common emitter small-signal short-circuit forward- current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 10 mA dc; f = 100 MHz	$\|h_{fe2}\|$	20		
Open capacitance (open circuit)	3236	V_{CB} = 4.0 V dc; I_E = 0; 100 kHz ≤ f ≤ 1.0 MHz	C_{obo}		2.5	pF
Input capacitance (output open circuit)	3240	V_{EB} = 0.5 V dc; I_C = 0; 100 kHz ≤ f ≤ 1.0 MHz	C_{ibo}		2.5	pF
Collector-base time constant		V_{CE} = 4.0 V dc; I_C = 5.0 mA dc; f = 31.8 MHz; see 4.5.2 and figure 2	$r'_b C_{c1}$		60	ps
Collector-base time constant		V_{CE} = 4.0 V dc; I_C = 10 mA dc; f = 31.8 MHz; see 4.5.2 and figure 2	$r'_b C_{c2}$		50	ps
Pulse response:	3251	Test condition B, except test circuit and pulse requirements in accordance with figure 3 herein.				
Turn-on time		V_{CC} = 17 V dc; I_C = 10 mA dc;	t_{on}		2.5	ns
Turn-off time		V_{CC} = 17 V dc; I_C = 10 mA dc;	t_{off}		3.5	ns
Subgroups 5, 6 and 7						
Not applicable						

1/ For sampling plan see MIL-PRF 19500.

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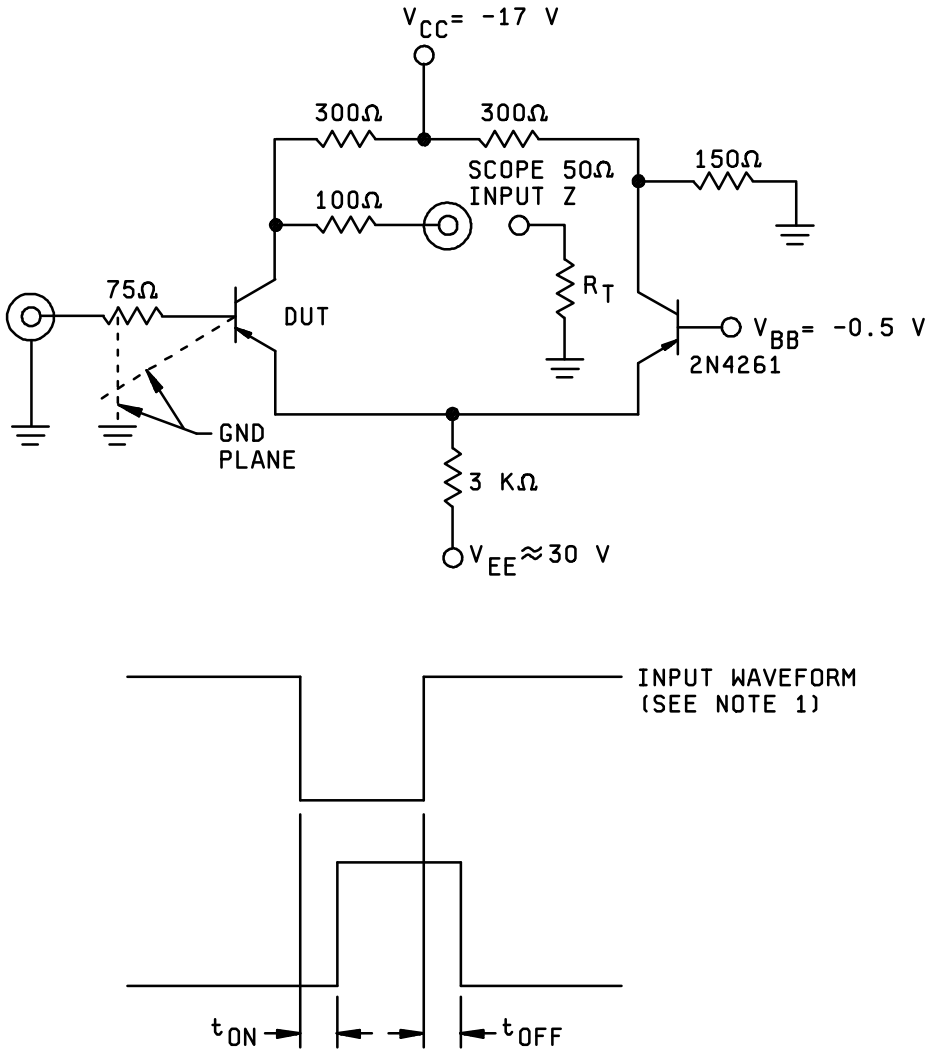
Procedure:

1. Set signal generator to 31.8 MHz and connect to “in” connector on test jig.
2. Connect low voltage dc power supplies as shown. A 1 k ohm resistor should be placed in series with the emitter power supply to prevent damage to transistors being tested.
3. Place transistor to be tested in socket. Set collector supply for V_{CE} = 4.0 V dc, and emitter supply for I_C as specified.
4. Connect RF voltmeter with unterminated probe adapter to “CAL” connector on test jig. Adjust signal generator until RF voltage is 0.5 volts. (Note: Decade switching of voltmeter should be accurate from 1 mV to 3 volts. If not, input voltage may be set using voltage dividers, utilizing lower scales of the RF voltmeter. If this is done, the voltage dividers should be left in place when the voltmeter is removed, as they constitute a load on the input of the circuit.
5. Remove RF voltmeter from “CAL” connector to “OUT” connector. Meter will now read r'_b, C_c as follows:

Meter range full scale	r'_b, C_c range
0.003 volts	10 to 30 ps
0.01 volts	30 to 100 ps
0.03 volts	100 to 300 ps

FIGURE 2. Collector base time constant test circuit (an equivalent circuit may be used).

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NOTES:
1. The input waveform is supplied by a pulse generator with the following characteristics: Z_{OUT} = 50 ohms; PW = 30 ns; duty cycle ≤ 2 percent.
2. Resistors shall be noninductive types.
3. The dc power supplies may require additional by passing in order to minimize ringing.

FIGURE 3. Switching time test circuit and waveforms.

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PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements should be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents see MIL-PRF-19500

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturer's List QML-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – CR
Air Force - 11

Review Activities:
Air Force – 13, 15, 19, 99

Preparing activity:
DLA-CC
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INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/511C	2. DOCUMENT DATE (YYMMDD) 990515

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, SWITCHING TYPE 2N4261 JAN, JANTX AND JANTXV AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN FAX EMAIL	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. Point of contact: Alan Barone,	b. TELEPHONE Commercial 614-692-0510 DSN 850-0510 FAX 614-692-6939 EMAIL alan_barone@dscc.dla.mil
c. ADDRESS : Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman, Suite 2533, Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-68880

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